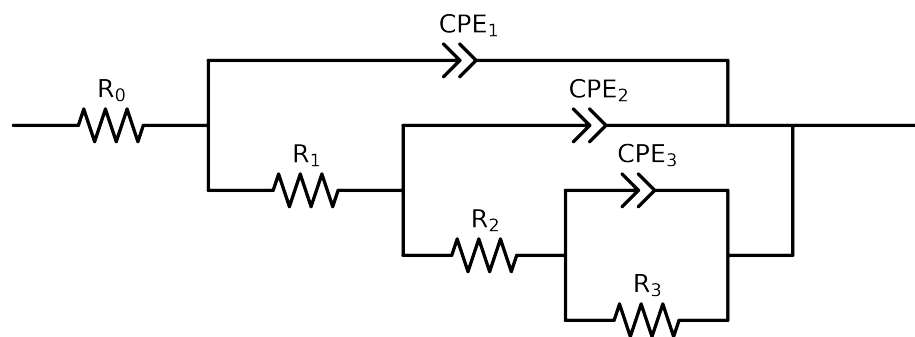


Comparative EIS Kinetics Report

Model Structure: $R_0 - p(R_1 - p(R_2 - p(R_3 - p(R_4, CPE_4), CPE_3), CPE_2), CPE_1)$

Used data: original data



Element	Unit	Bounds	Vasco_ LNV3_ 4	Vasco_ LNV3_ 5	Vasco_ LNV3_ 7d_
R_0	Ω	$[1.0e+00, 1.0e+03]$	$1.56e+02 \pm 1.9e+00$	$1.25e+02 \pm 2.5e+00$	$1.06e+02 \pm 2.2e+00$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$4.69e+02 \pm 4.8e+01$	$8.28e+02 \pm 7.0e+01$	$9.38e+02 \pm 7.6e+01$
R_2	Ω	$[1.0e+00, 1.0e+08]$	$2.64e+05 \pm 6.6e+04$	$4.19e+05 \pm 7.0e+04$	$2.14e+03 \pm 8.0e+02$
R_3	Ω	$[1.0e+00, 1.0e+08]$	$8.33e+05 \pm 1.3e+11$	$6.01e+05 \pm 2.2e+07$	$8.53e+05 \pm 8.8e+07$
R_4	Ω	$[1.0e+00, 1.0e+08]$	$5.57e+04 \pm 1.3e+11$	$7.47e+00 \pm 2.2e+07$	$3.91e+01 \pm 8.8e+07$
Q_4	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$2.08e-07 \pm 1.2e+00$	$1.18e-07 \pm 4.2e-10$	$3.11e-05 \pm 1.7e+02$
n_4	-	$[5.0e-01, 1.0e+00]$	$6.39e-01 \pm 2.7e+05$	$9.01e-01 \pm 2.3e-03$	$9.03e-01 \pm 6.7e+05$
Q_3	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$1.70e-05 \pm 2.7e-03$	$4.55e-05 \pm 2.9e-05$	$2.78e-06 \pm 1.2e-06$
n_3	-	$[5.0e-01, 1.0e+00]$	$8.24e-01 \pm 1.9e+01$	$1.00e+00 \pm 2.2e-01$	$1.00e+00 \pm 5.3e-02$
Q_2	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-02]$	$3.67e-06 \pm 6.5e-07$	$5.96e-06 \pm 8.3e-07$	$1.30e-06 \pm 1.2e-06$
n_2	-	$[5.0e-01, 1.0e+00]$	$9.12e-01 \pm 2.1e-02$	$8.75e-01 \pm 2.1e-02$	$1.00e+00 \pm 1.4e-01$
Q_1	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$3.63e-06 \pm 9.5e-07$	$5.85e-06 \pm 1.1e-06$	$1.18e-05 \pm 3.4e-07$
n_1	-	$[5.0e-01, 1.0e+00]$	$7.72e-01 \pm 2.6e-02$	$6.87e-01 \pm 1.9e-02$	$5.91e-01 \pm 3.1e-03$
χ^2	-	-	$8.240e-04$	$1.513e-03$	$1.172e-03$

Analysis Results: Vasco_LNV3_4

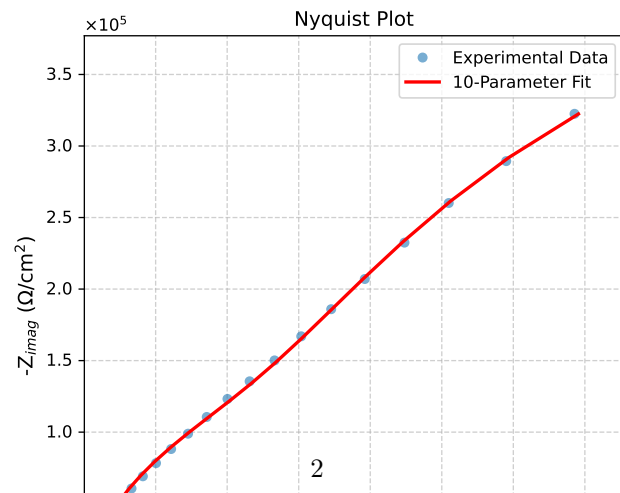


Figure 1: EIS Fit Results for Vasco.LNV3.4

Fitted Data: Vasco_LNV3_4

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0008e+05	1.5830e+02	8.4945e+00	1.5952e+02	-3.0524
7.9453e+04	1.5998e+02	1.0123e+01	1.6030e+02	-3.6205
6.3141e+04	1.6081e+02	1.2047e+01	1.6126e+02	-4.2843
5.0203e+04	1.6181e+02	1.4321e+01	1.6244e+02	-5.0580
3.9891e+04	1.6303e+02	1.7019e+01	1.6392e+02	-5.9596
3.1641e+04	1.6454e+02	2.0230e+01	1.6578e+02	-7.0095
2.5172e+04	1.6636e+02	2.3963e+01	1.6807e+02	-8.1968
2.0016e+04	1.6859e+02	2.8350e+01	1.7096e+02	-9.5455
1.5891e+04	1.7136e+02	3.3513e+01	1.7461e+02	-11.0658
1.2609e+04	1.7478e+02	3.9534e+01	1.7920e+02	-12.7453
1.0078e+04	1.7887e+02	4.6264e+01	1.8475e+02	-14.5018
8.0156e+03	1.8400e+02	5.4129e+01	1.9179e+02	-16.3931
6.3281e+03	1.9054e+02	6.3355e+01	2.0080e+02	-18.3919
5.0156e+03	1.9846e+02	7.3534e+01	2.1165e+02	-20.3306
3.9844e+03	2.0802e+02	8.4646e+01	2.2459e+02	-22.1417
3.1710e+03	2.1944e+02	9.6557e+01	2.3974e+02	-23.7502
2.5276e+03	2.3286e+02	1.0905e+02	2.5713e+02	-25.0941
1.9761e+03	2.4984e+02	1.2304e+02	2.7849e+02	-26.2183
1.5775e+03	2.6746e+02	1.3591e+02	3.0001e+02	-26.9381
1.2656e+03	2.8626e+02	1.4836e+02	3.2242e+02	-27.3959
9.9826e+02	3.0767e+02	1.6160e+02	3.4753e+02	-27.7092
7.9688e+02	3.2838e+02	1.7432e+02	3.7179e+02	-27.9615
6.2779e+02	3.4991e+02	1.8878e+02	3.9759e+02	-28.3477
5.0551e+02	3.6856e+02	2.0393e+02	4.2121e+02	-28.9561
3.9800e+02	3.8781e+02	2.2453e+02	4.4812e+02	-30.0690
3.1550e+02	4.0520e+02	2.5039e+02	4.7632e+02	-31.7138
2.5240e+02	4.2103e+02	2.8271e+02	5.0714e+02	-33.8796
1.9862e+02	4.3774e+02	3.2795e+02	5.4696e+02	-36.8404
1.5836e+02	4.5406e+02	3.8325e+02	5.9419e+02	-40.1662
1.2556e+02	4.7231e+02	4.5541e+02	6.5611e+02	-43.9559
1.0045e+02	4.9235e+02	5.4263e+02	7.3270e+02	-47.7812
7.9003e+01	5.1794e+02	6.6012e+02	8.3906e+02	-51.8815
6.3345e+01	5.4652e+02	7.9422e+02	9.6409e+02	-55.4674
5.0223e+01	5.8339e+02	9.6760e+02	1.1299e+03	-58.9132
3.8422e+01	6.3725e+02	1.2181e+03	1.3747e+03	-62.3827
3.1250e+01	6.8936e+02	1.4561e+03	1.6110e+03	-64.6649
2.4934e+01	7.5962e+02	1.7703e+03	1.9264e+03	-66.7762
1.9862e+01	8.4785e+02	2.1554e+03	2.3161e+03	-68.5268
1.5625e+01	9.6492e+02	2.6522e+03	2.8223e+03	-70.0076
1.2401e+01	1.1070e+03	3.2373e+03	3.4213e+03	-71.1209
9.9311e+00	1.2779e+03	3.9187e+03	4.1218e+03	-71.9383
7.9449e+00	1.4923e+03	4.7450e+03	4.9741e+03	-72.5411
6.3174e+00	1.7690e+03	5.7711e+03	6.0362e+03	-72.9587
5.0080e+00	2.1236e+03	7.0313e+03	7.3450e+03	-73.1946
3.9457e+00	2.5878e+03	8.6022e+03	8.9830e+03	-73.2571
3.1587e+00	3.1391e+03	1.0371e+04	1.0836e+04	-73.1602
2.5040e+00	3.8723e+03	1.2590e+04	1.3172e+04	-72.9031
1.9981e+00	4.7845e+03	1.5174e+04	1.5911e+04	-72.4995
1.5847e+00	5.9879e+03	1.8343e+04	1.9296e+04	-71.9216
1.2669e+00	7.4822e+03	2.1976e+04	2.3215e+04	-71.1977
9.9904e-01	9.5322e+03	2.6533e+04	2.8193e+04	-70.2386
7.9234e-01	1.2128e+04	3.1758e+04	3.3995e+04	-69.0993
6.3345e-01	1.5345e+04	3.7596e+04	4.0607e+04	-67.7967
5.0403e-01	1.9531e+04	4.4395e+04	4.8501e+04	-66.2540
4.0064e-01	2.4849e+04	5.2071e+04	5.7697e+04	-64.4892
3.1672e-01	3.1653e+04	6.0743e+04	6.8495e+04	-62.4758
2.5202e-01	3.9744e+04	6.9836e+04	8.0354e+04	-60.3554
2.0032e-01	4.9411e+04	7.9465e+04	9.3574e+04	-58.1269
1.5890e-01	6.0685e+04	8.9524e+04	1.0815e+05	-55.8681
1.2601e-01	7.3340e+04	9.9869e+04	1.2391e+05	-53.7078
1.0016e-01	8.6986e+04	1.1047e+05	1.4061e+05	-51.7825
7.9449e-02	1.0168e+05	1.2183e+05	1.5869e+05	-50.1523
6.3174e-02	1.1707e+05	1.3422e+05	1.7810e+05	-48.9048
5.0134e-02	1.3362e+05	1.4848e+05	1.9975e+05	-48.0157
3.9826e-02	1.5157e+05	1.6506e+05	2.2409e+05	-47.4389
3.1630e-02	1.7175e+05	1.8454e+05	2.5210e+05	-47.0567
2.5121e-02	1.9512e+05	2.0723e+05	2.8463e+05	-46.7229
1.9955e-02	2.2287e+05	2.3307e+05	3.2248e+05	-46.2809
1.5852e-02	2.5635e+05	2.6165e+05	3.6629e+05	-45.5861
1.2591e-02	2.9688e+05	2.9198e+05	4.1640e+05	-44.5231
1.0001e-02	3.4550e+05	3.2240e+05	4.7256e+05	-43.0191

Analysis Results: Vasco_LNV3_5

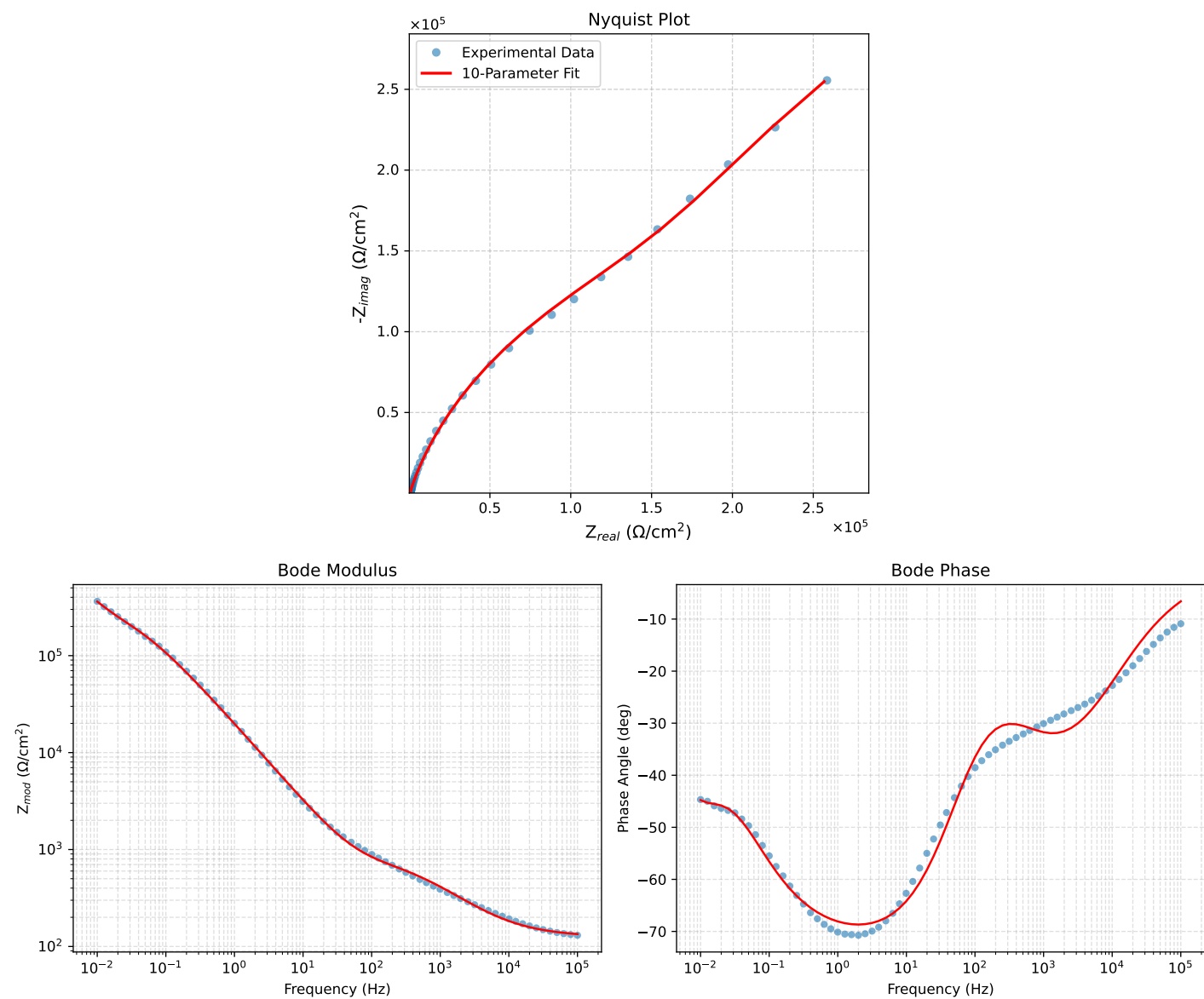


Figure 2: EIS Fit Results for Vasco_LNV3_5

Fitted Data: Vasco_LNV3_5

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^{\circ}$)
1.0008e+05	1.3338e+02	1.5402e+01	1.3427e+02	-6.5871
7.9453e+04	1.3490e+02	1.7982e+01	1.3609e+02	-7.5927
6.3141e+04	1.3669e+02	2.0967e+01	1.3829e+02	-8.7209
5.0203e+04	1.3880e+02	2.4422e+01	1.4093e+02	-9.9793
3.9891e+04	1.4129e+02	2.8432e+01	1.4412e+02	-11.3777
3.1641e+04	1.4427e+02	3.3105e+01	1.4802e+02	-12.9236
2.5172e+04	1.4776e+02	3.8423e+01	1.5267e+02	-14.5765
2.0016e+04	1.5190e+02	4.4545e+01	1.5830e+02	-16.3442
1.5891e+04	1.5685e+02	5.1605e+01	1.6512e+02	-18.2121
1.2609e+04	1.6274e+02	5.9677e+01	1.7334e+02	-20.1383
1.0078e+04	1.6951e+02	6.8538e+01	1.8284e+02	-22.0151
8.0156e+03	1.7770e+02	7.8728e+01	1.9436e+02	-23.8949
6.3281e+03	1.8775e+02	9.0515e+01	2.0843e+02	-25.7382
5.0156e+03	1.9949e+02	1.0339e+02	2.2469e+02	-27.3962
3.9844e+03	2.1320e+02	1.1737e+02	2.4338e+02	-28.8339
3.1710e+03	2.2915e+02	1.3238e+02	2.6464e+02	-30.0159
2.5276e+03	2.4755e+02	1.4826e+02	2.8855e+02	-30.9178
1.9761e+03	2.7065e+02	1.6632e+02	3.1767e+02	-31.5715
1.5775e+03	2.9477e+02	1.8330e+02	3.4711e+02	-31.8746
1.2656e+03	3.2107e+02	1.9999e+02	3.7826e+02	-31.9179
9.9826e+02	3.5217e+02	2.1774e+02	4.1404e+02	-31.7277
7.9688e+02	3.8391e+02	2.3416e+02	4.4969e+02	-31.3809
6.2779e+02	4.1916e+02	2.5111e+02	4.8862e+02	-30.9252
5.0551e+02	4.5191e+02	2.6644e+02	5.2461e+02	-30.5230
3.9800e+02	4.8803e+02	2.8411e+02	5.6470e+02	-30.2061
3.1550e+02	5.2231e+02	3.0331e+02	6.0399e+02	-30.1438
2.5240e+02	5.5403e+02	3.2528e+02	6.4247e+02	-30.4180
1.9862e+02	5.8663e+02	3.5496e+02	6.8566e+02	-31.1775
1.5836e+02	6.1626e+02	3.9121e+02	7.2995e+02	-32.4078
1.2556e+02	6.4606e+02	4.3937e+02	7.8131e+02	-34.2184
1.0045e+02	6.7506e+02	4.9898e+02	8.3945e+02	-36.4702
7.9003e+01	7.0796e+02	5.8118e+02	9.1596e+02	-39.3830
6.3345e+01	7.4122e+02	6.7682e+02	1.0037e+03	-42.3997
5.0223e+01	7.8100e+02	8.0218e+02	1.1196e+03	-45.7667
3.8422e+01	8.3576e+02	9.8502e+02	1.2918e+03	-49.6864
3.1250e+01	8.8673e+02	1.1595e+03	1.4597e+03	-52.5932
2.4934e+01	9.5375e+02	1.3900e+03	1.6858e+03	-55.5445
1.9862e+01	1.0362e+03	1.6719e+03	1.9669e+03	-58.2092
1.5625e+01	1.1437e+03	2.0338e+03	2.3334e+03	-60.6485
1.2401e+01	1.2721e+03	2.4574e+03	2.7671e+03	-62.6313
9.9311e+00	1.4239e+03	2.9472e+03	3.2731e+03	-64.2133
7.9449e+00	1.6109e+03	3.5363e+03	3.8859e+03	-65.5092
6.3174e+00	1.8473e+03	4.2618e+03	4.6449e+03	-66.5651
5.0080e+00	2.1433e+03	5.1448e+03	5.5734e+03	-67.3836
3.9457e+00	2.5203e+03	6.2357e+03	6.7258e+03	-67.9928
3.1587e+00	2.9550e+03	7.4542e+03	8.0185e+03	-68.3758
2.5040e+00	3.5146e+03	8.9711e+03	9.6350e+03	-68.6062
1.9981e+00	4.1867e+03	1.0728e+04	1.1516e+04	-68.6804
1.5847e+00	5.0408e+03	1.2873e+04	1.3824e+04	-68.6152
1.2669e+00	6.0613e+03	1.5328e+04	1.6483e+04	-68.4236
9.9904e-01	7.4077e+03	1.8414e+04	1.9848e+04	-68.0861
7.9234e-01	9.0494e+03	2.1981e+04	2.3771e+04	-67.6232
6.3345e-01	1.1020e+04	2.6021e+04	2.8258e+04	-67.0479
5.0403e-01	1.3519e+04	3.0832e+04	3.3665e+04	-66.3239
4.0064e-01	1.6648e+04	3.6444e+04	4.0067e+04	-65.4489
3.1672e-01	2.0650e+04	4.3079e+04	4.7772e+04	-64.3892
2.5202e-01	2.5496e+04	5.0451e+04	5.6527e+04	-63.1893
2.0032e-01	3.1518e+04	5.8798e+04	6.6712e+04	-61.8072
1.5890e-01	3.8987e+04	6.8158e+04	7.8521e+04	-60.2300
1.2601e-01	4.8085e+04	7.8389e+04	9.1962e+04	-58.4742
1.0016e-01	5.8859e+04	8.9212e+04	1.0688e+05	-56.5845
7.9449e-02	7.1590e+04	1.0063e+05	1.2350e+05	-54.5707
6.3174e-02	8.5958e+04	1.1221e+05	1.4135e+05	-52.5464
5.0134e-02	1.0199e+05	1.2407e+05	1.6061e+05	-50.5785
3.9826e-02	1.1913e+05	1.3618e+05	1.8094e+05	-48.8217
3.1630e-02	1.3712e+05	1.4909e+05	2.0255e+05	-47.3947
2.5121e-02	1.5585e+05	1.6360e+05	2.2595e+05	-46.3890
1.9955e-02	1.7574e+05	1.8075e+05	2.5211e+05	-45.8048
1.5852e-02	1.9793e+05	2.0152e+05	2.8246e+05	-45.5150
1.2591e-02	2.2421e+05	2.2636e+05	3.1860e+05	-45.2744
1.0001e-02	2.5674e+05	2.5479e+05	3.6171e+05	-44.7811

Analysis Results: Vasco_LNV3_7d_

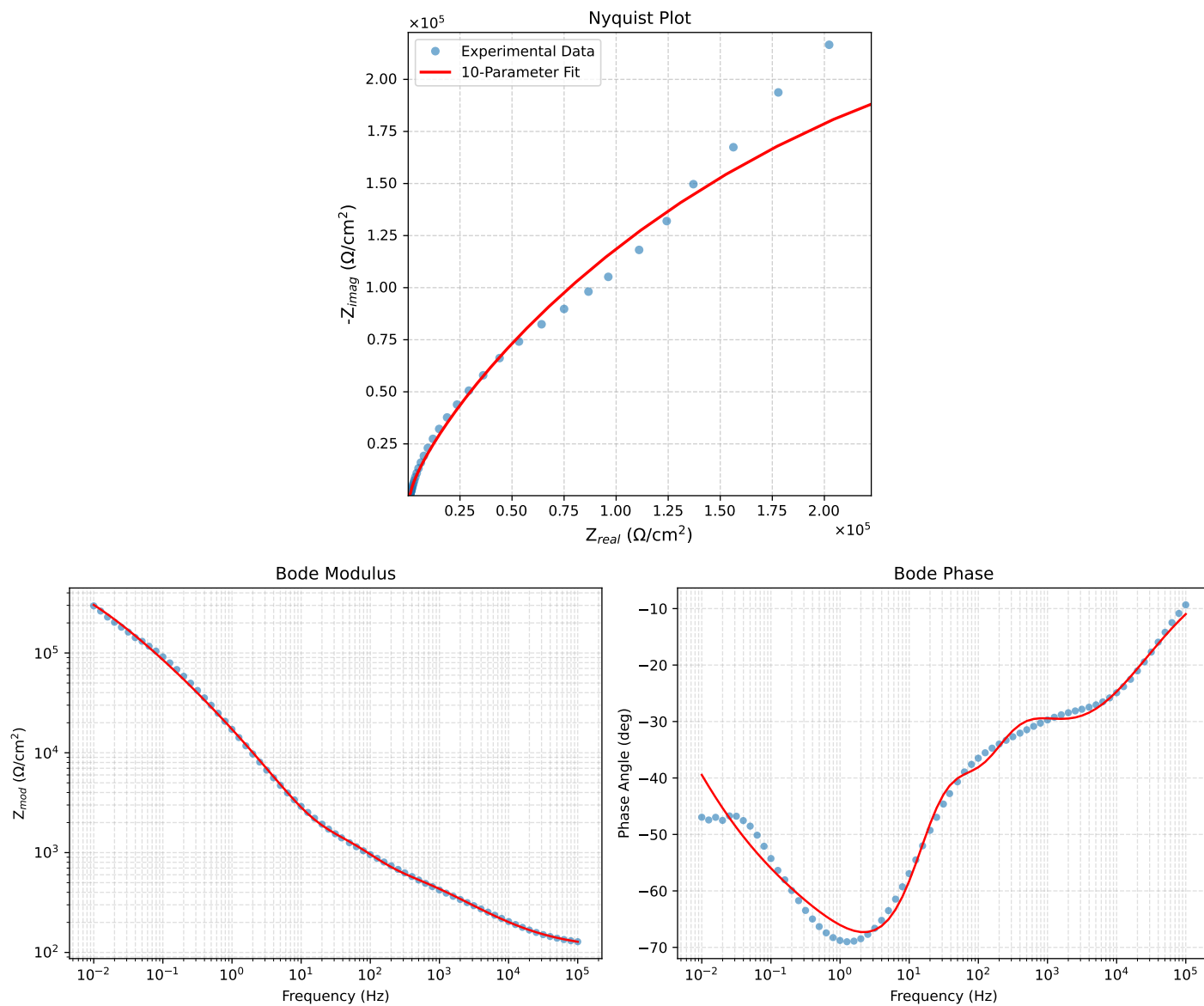


Figure 3: EIS Fit Results for Vasco_LNV3_7d_

Fitted Data: Vasco_LNV3_7d_

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^{\circ}$)
1.0002e+05	1.2566e+02	2.4367e+01	1.2800e+02	-10.9743
7.9512e+04	1.2849e+02	2.7740e+01	1.3145e+02	-12.1826
6.3105e+04	1.3178e+02	3.1582e+01	1.3551e+02	-13.4775
5.0215e+04	1.3550e+02	3.5866e+01	1.4017e+02	-14.8256
3.9902e+04	1.3981e+02	4.0716e+01	1.4562e+02	-16.2366
3.1699e+04	1.4477e+02	4.6166e+01	1.5195e+02	-17.6873
2.5137e+04	1.5051e+02	5.2316e+01	1.5935e+02	-19.1667
1.9980e+04	1.5704e+02	5.9103e+01	1.6780e+02	-20.6237
1.5879e+04	1.6454e+02	6.6642e+01	1.7753e+02	-22.0484
1.2598e+04	1.7321e+02	7.5032e+01	1.8876e+02	-23.4217
1.0020e+04	1.8302e+02	8.4146e+01	2.0144e+02	-24.6911
7.9282e+03	1.9449e+02	9.4314e+01	2.1615e+02	-25.8704
6.3079e+03	2.0725e+02	1.0507e+02	2.3236e+02	-26.8830
5.0347e+03	2.2149e+02	1.1643e+02	2.5023e+02	-27.7287
3.9931e+03	2.3800e+02	1.2883e+02	2.7063e+02	-28.4273
3.1441e+03	2.5712e+02	1.4229e+02	2.9387e+02	-28.9604
2.5195e+03	2.7679e+02	1.5525e+02	3.1736e+02	-29.2882
2.0008e+03	2.9925e+02	1.6913e+02	3.4373e+02	-29.4743
1.5807e+03	3.2416e+02	1.8362e+02	3.7256e+02	-29.5295
1.2536e+03	3.5035e+02	1.9818e+02	4.0252e+02	-29.4951
1.0007e+03	3.7706e+02	2.1277e+02	4.3295e+02	-29.4355
7.9003e+02	4.0595e+02	2.2897e+02	4.6607e+02	-29.4247
6.2881e+02	4.3427e+02	2.4625e+02	4.9923e+02	-29.5546
5.0223e+02	4.6236e+02	2.6594e+02	5.3339e+02	-29.9071
4.0015e+02	4.9110e+02	2.9003e+02	5.7035e+02	-30.5644
3.1550e+02	5.2238e+02	3.2151e+02	6.1339e+02	-31.6112
2.5202e+02	5.5458e+02	3.5876e+02	6.6051e+02	-32.8989
2.0032e+02	5.9230e+02	4.0534e+02	7.1772e+02	-34.3860
1.5801e+02	6.3895e+02	4.6256e+02	7.8881e+02	-35.9022
1.2556e+02	6.9354e+02	5.2563e+02	8.7022e+02	-37.1584
9.9734e+01	7.5800e+02	5.9447e+02	9.6330e+02	-38.1056
7.9449e+01	8.2977e+02	6.6647e+02	1.0643e+03	-38.7713
6.2919e+01	9.0829e+02	7.4462e+02	1.1745e+03	-39.3451
4.9867e+01	9.8694e+02	8.2993e+02	1.2895e+03	-40.0609
3.8422e+01	1.0707e+03	9.4267e+02	1.4265e+03	-41.3617
3.1250e+01	1.1319e+03	1.0535e+03	1.5463e+03	-42.9454
2.4934e+01	1.1943e+03	1.2070e+03	1.6980e+03	-45.3018
2.0032e+01	1.2532e+03	1.3987e+03	1.8780e+03	-48.1395
1.5625e+01	1.3246e+03	1.6819e+03	2.1408e+03	-51.7781
1.2467e+01	1.4012e+03	2.0133e+03	2.4529e+03	-55.1630
9.9734e+00	1.4968e+03	2.4234e+03	2.8484e+03	-58.2984
7.9719e+00	1.6225e+03	2.9315e+03	3.3506e+03	-61.0364
6.3516e+00	1.7920e+03	3.5611e+03	3.9866e+03	-63.2881
5.0134e+00	2.0277e+03	4.3578e+03	4.8065e+03	-65.0472
3.9860e+00	2.3302e+03	5.2882e+03	5.7788e+03	-66.2200
3.1758e+00	2.7199e+03	6.3863e+03	6.9413e+03	-66.9313
2.5161e+00	3.2339e+03	7.7200e+03	8.3700e+03	-67.2709
1.9955e+00	3.8866e+03	9.2870e+03	1.0067e+04	-67.2906
1.5879e+00	4.6967e+03	1.1098e+04	1.2051e+04	-67.0618
1.2581e+00	5.7273e+03	1.3253e+04	1.4437e+04	-66.6281
9.9777e-01	6.9993e+03	1.5749e+04	1.7234e+04	-66.0384
7.9274e-01	8.5507e+03	1.8617e+04	2.0487e+04	-65.3310
6.2903e-01	1.0456e+04	2.1943e+04	2.4307e+04	-64.5222
4.9888e-01	1.2775e+04	2.5773e+04	2.8766e+04	-63.6333
3.9860e-01	1.5481e+04	3.0009e+04	3.3767e+04	-62.7109
3.1672e-01	1.8805e+04	3.4943e+04	3.9682e+04	-61.7119
2.5148e-01	2.2797e+04	4.0557e+04	4.6525e+04	-60.6603
1.9998e-01	2.7521e+04	4.6850e+04	5.4335e+04	-59.5693
1.5879e-01	3.3165e+04	5.3961e+04	6.3338e+04	-58.4245
1.2601e-01	3.9865e+04	6.1924e+04	7.3646e+04	-57.2276
1.0016e-01	4.7707e+04	7.0693e+04	8.5284e+04	-55.9867
7.9503e-02	5.6971e+04	8.0406e+04	9.8543e+04	-54.6805
6.3140e-02	6.7785e+04	9.0985e+04	1.1346e+05	-53.3134
5.0123e-02	8.0401e+04	1.0244e+05	1.3022e+05	-51.8733
3.9833e-02	9.4945e+04	1.1462e+05	1.4884e+05	-50.3634
3.1638e-02	1.1173e+05	1.2748e+05	1.6951e+05	-48.7674
2.5126e-02	1.3094e+05	1.4082e+05	1.9229e+05	-47.0830
1.9957e-02	1.5271e+05	1.5437e+05	2.1715e+05	-45.3090
1.5849e-02	1.7722e+05	1.6783e+05	2.4408e+05	-43.4425
1.2590e-02	2.0444e+05	1.8081e+05	2.7293e+05	-41.4893
1.0001e-02	2.3438e+05	1.9288e+05	3.0354e+05	-39.4525